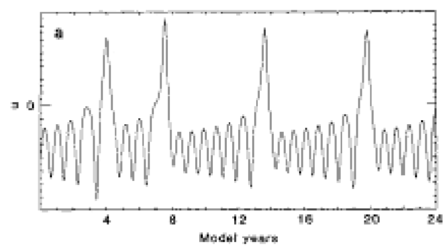


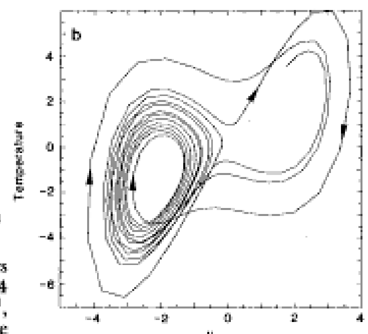
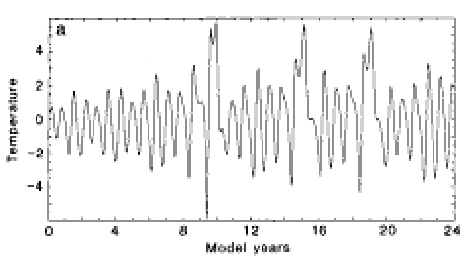




Fig. 3. (a) Typical model time series of $u$. Parameters chosen correspond to frictional time scales ($C$) of 1/4 month$^{-1}$, temperature decay time scale ($A$) of 1 year$^{-1}$, mean forcing current ($u^*$) of $-0.45$ m sec$^{-1}$, basin size ($\Delta x$) of 7500 km, and $T^*$ of 12°C. $A$ and $C$ are similar to those of many modeling studies (*6, 8*). The parameter $u^*$ has the value of typically observed mean near-surface westward currents (*17*). $T^*$ is a typical temperature difference across the thermocline (*18*). A realistic value of $B$ is difficult to assess. The value used here (2 m$^2$ sec$^{-2}$ C$^{-1}$) is approximately equivalent to supposing a current of 35 cm sec$^{-1}$ can be generated in 1 month by a temperature difference $(T_e - T_w)$ of 2°C. (b) Trajectory of system projected onto $u$, $(T_e - T_w)$ plane, over a period of 12 years. Arrows indicate direction of flow. Model has previously been integrated for several years, and the initial conditions "forgotten." Time-stepping uses fourth-order Runge-Kutta.



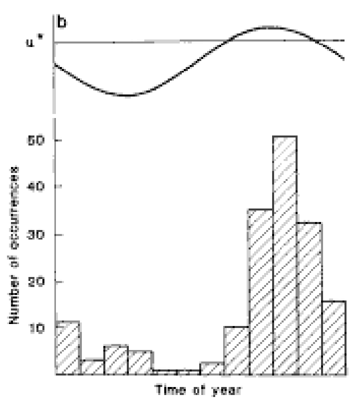




Fig. 4. (a) Model time series of $T_e - T_w$ with seasonal cycle modeled by allowing $u^*$ to vary as $u^*$ (1 + 3 sin $\omega t$). Other parameters are unchanged. Here, the model variability is not completely dominated by the seasonal cycle. However, the onset of El Niño occurs only during part of the annual cycle, as indicated in (b), a histogram of occurrences of model El Niño onset (defined by $u$ becoming positive) over time of year for a 960-year integration. The upper part of (b) illustrates the variability of $u^*$ over the year.

apparently randomly between the two least unstable stationary states (the solution with $u = 0$ is most unstable and is essentially ignored). With $u^* \neq 0$, the symmetry between $T_e$ and $T_w$ is broken. Numerical time integration of the equations then yields aperiodic behavior remarkably evocative of El

event cycle can plausibly be reproduced. After a large model El Niño, the system returns close to the other stationary solution. It then takes longer before the model breaks away to another El Niño, in accordance with observations (*1*).

A seasonal cycle may be introduced by

plus seasonal forcing. However, no such behavior was found.] Finally, the basic size may be altered by varying $\Delta x$ in Eqs. 1 to 3. The temperature difference between normal and El Niño states is then altered almost in direct proportion. Thus, reducing $\Delta x$ by a factor of 3 (for the Atlantic Ocean) markedly reduces the El Niño effect although the associated winds are stronger.

In summary, although the model presented here has many limitations (one being that it cannot describe spatial variations in any detail), it does explain many of the qualitative features of El Niño. It transparently demonstrates the underlying dynamics and thereby the possibility of a purely internal mechanism for the phenomenon. It shows that external triggering or stochastic forcing is not necessarily essential, although such effects may have a role in the real system. Verification or falsification of such a theory of El Niño will probably arise through controlled numerical experimentation involving large coupled general circulation models that can be directly compared with observations and used to judge a simpler theory. Until then, the theory presented here should be regarded as untested.

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Niño (Fig. 3). The system stays in the neighborhood of one stationary (but unstable) solution of the equations for many model years before flipping to an El Niño state. El Niño states are orbits around another stationary, but highly unstable, solution to the equations and the system quickly returns to its normal, less unstable state. Model El Niños occur aperiodically, but typically with an interval of a few to several model years. Thus, with neither seasonal cycling nor stochastic forcing, the basic

varying $u^*$, in order to represent an annual cycle of the trade winds. (The parameter $T^*$ may also be varied, with a similar effect.) Numerical integrations then reveal that the onset of El Niño is partially "phase-locked" to the annual cycle (Fig. 4). Only when the trade winds are weak is the model able to break into an El Niño. This is a well-known feature of the real system. [It may be theoretically possible to produce chaotic behavior with a two-variable model, say with temperature difference $(T_e - T_w)$ and $u$,

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